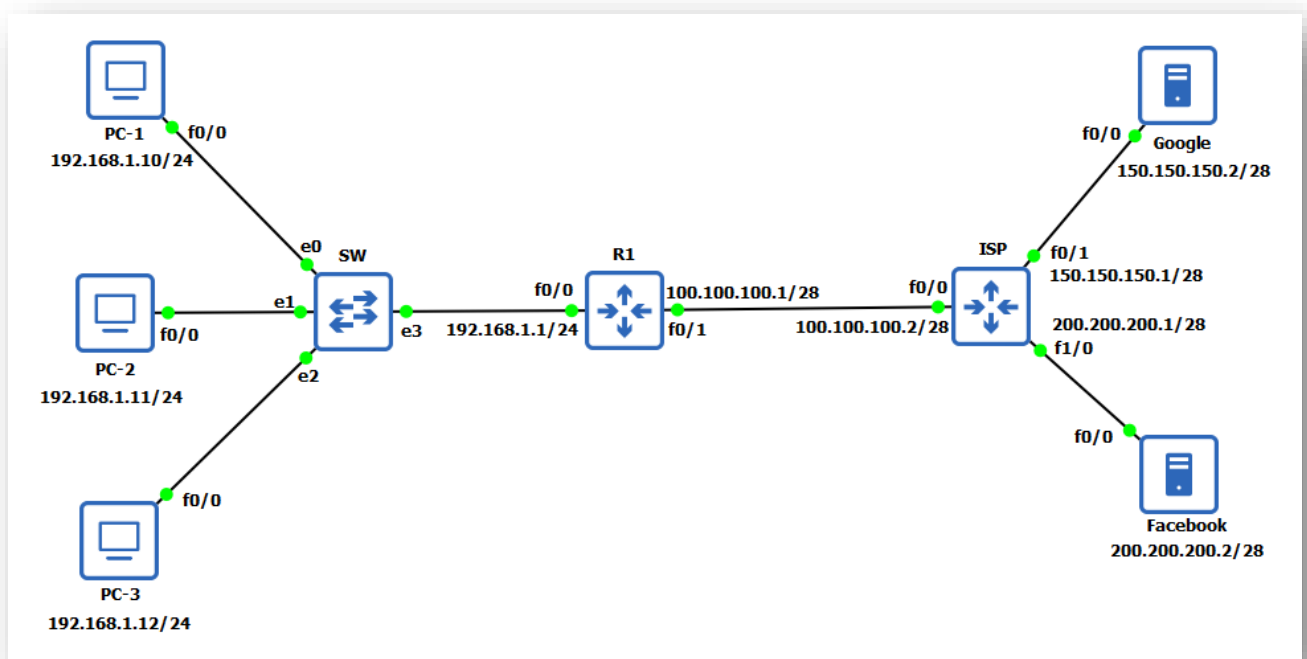


NAT Overload (PAT)



Task

1. Configure routers R1 & ISP with IP address as shown in topology and configure enable password as **ccna**. (**Refer Lab-19 for task-1 to task-4**)
2. Configure routers as Host (PC)
3. Configure routers as Host (Servers) and enable HTTP service on it.
4. Configure default routing on R1.
5. Configure NAT overload with R1's WAN interface IP for 192.168.1.0/24 network.
6. Configure NAT overload with IP Pool for 192.168.1.0/24 network.

Task-5: Configure NAT overload.

```

R1# config t
R1(config)#access-list 10 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface fa0/1 overload
R1(config)#int fa0/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#

```

✓ **Verification & Testing:**

R1#sh ip nat translations

Enable icmp debugging on Google server, Facebook server and PC-1. Ping Google and Facebook server from PC-1, PC-2 and PC-3

Google#debug ip icmp

ICMP packet debugging is on

Google#

Facebook#debug ip icmp

ICMP packet debugging is on

Facebook#

PC-1#debug ip icmp

ICMP packet debugging is on

PC-1#

PC-1#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/59/72 ms

PC-1#

PC-1#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/56/64 ms

PC-1#

PC-1#telnet 150.150.150.2 80

Trying 150.150.150.2, 80 ... Open

PC-1#

PC-1#telnet 200.200.200.2 80

Trying 200.200.200.2, 80 ... Open

PC-1#

PC-2#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 56/62/80 ms

PC-2#

PC-2#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/54/60 ms

PC-2#

PC-2#telnet 150.150.150.2 80

Trying 150.150.150.2, 80 ... Open

PC-2#

PC-2#telnet 200.200.200.2 80

Trying 200.200.200.2, 80 ... Open

PC-2#

PC-3#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/54/72 ms

PC-3#

PC-3#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/55/68 ms

PC-3#

PC-3#telnet 150.150.150.2 80

Trying 150.150.150.2, 80 ... Open

PC-3#

PC-3#telnet 200.200.200.2 80

Trying 200.200.200.2, 80 ... Open

PC-3#

R1#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	100.100.100.1:4	192.168.1.10:4	150.150.150.2:4	150.150.150.2:4
icmp	100.100.100.1:5	192.168.1.10:5	200.200.200.2:5	200.200.200.2:5
tcp	100.100.100.1:41224	192.168.1.10:41224	150.150.150.2:80	150.150.150.2:80
tcp	100.100.100.1:61810	192.168.1.10:61810	200.200.200.2:80	200.200.200.2:80
icmp	100.100.100.1:0	192.168.1.11:4	150.150.150.2:4	150.150.150.2:0
icmp	100.100.100.1:1	192.168.1.11:5	200.200.200.2:5	200.200.200.2:1
tcp	100.100.100.1:22341	192.168.1.11:22341	200.200.200.2:80	200.200.200.2:80
tcp	100.100.100.1:54404	192.168.1.11:54404	150.150.150.2:80	150.150.150.2:80

```
icmp 100.100.100.1:2      192.168.1.12:2      150.150.150.2:2      150.150.150.2:2
icmp 100.100.100.1:3      192.168.1.12:3      200.200.200.2:3      200.200.200.2:3
tcp 100.100.100.1:29393 192.168.1.12:29393 200.200.200.2:80      200.200.200.2:80
tcp 100.100.100.1:30721 192.168.1.12:30721 150.150.150.2:80      150.150.150.2:80
R1#
```

Google#

```
*Nov 15 15:58:53.179: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:53.223: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:53.291: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:53.355: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:53.415: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
Google#
```

PC-1#

```
*Nov 15 16:02:15.823: ICMP: echo reply rcvd, src 150.150.150.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
*Nov 15 16:02:15.891: ICMP: echo reply rcvd, src 150.150.150.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
*Nov 15 16:02:15.955: ICMP: echo reply rcvd, src 150.150.150.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
*Nov 15 16:02:16.019: ICMP: echo reply rcvd, src 150.150.150.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
*Nov 15 16:02:16.087: ICMP: echo reply rcvd, src 150.150.150.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
PC-1#
```

Facebook#

```
*Nov 15 15:58:56.707: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:56.767: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:56.827: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:56.895: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
*Nov 15 15:58:56.959: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.1, topology
BASE, dscp 0 topoid 0
Facebook#
```

PC-1#

```
*Nov 15 16:03:21.671: ICMP: echo reply rcvd, src 200.200.200.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
*Nov 15 16:03:21.735: ICMP: echo reply rcvd, src 200.200.200.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
```

```
*Nov 15 16:03:21.795: ICMP: echo reply rcvd, src 200.200.200.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
*Nov 15 16:03:21.859: ICMP: echo reply rcvd, src 200.200.200.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
*Nov 15 16:03:21.919: ICMP: echo reply rcvd, src 200.200.200.2, dst 192.168.1.10, topology
BASE, dscp 0 topoid 0
PC-1#
```

R1#sh ip nat statistics

```
Total active translations: 6 (0 static, 6 dynamic; 6 extended)
Peak translations: 7, occurred 00:38:44 ago
Outside interfaces:
  FastEthernet0/1
Inside interfaces:
  FastEthernet0/0
Hits: 208 Misses: 0
CEF Translated packets: 208, CEF Punted packets: 0
Expired translations: 13
Dynamic mappings:
-- Inside Source
[Id: 2] access-list 10 interface FastEthernet0/1 refcount 6
Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
```

R1#

To Remove NAT overload.

```
R1#config t
R1(config)#no access-list 10 permit 192.168.1.0 0.0.0.255
R1(config)#no ip nat inside source list 10 interface fa0/1 overload
R1(config)#int fa0/0
R1(config-if)#no ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#no ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
```

Task-6: Configure NAT overload with IP Pool

```
R1#config t
R1(config)#access-list 10 permit 192.168.1.0 0.0.0.255
R1(config)#ip nat pool Public-IP 100.100.100.3 100.100.100.3 prefix-length 28
R1(config)#ip nat inside source list 10 pool Public-IP overload
R1(config)#int fa0/0
R1(config-if)#ip nat inside
R1(config-if)#exit
```

```
R1(config)#int fa0/1
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
```

✓ Verification & Testing:

R1#sh ip nat translations

Ping Google and Facebook server from PC-1, PC-2 and PC-3

PC-1#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 60/66/88 ms

PC-1#

PC-1#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/59/68 ms

PC-1#

PC-2#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/57/68 ms

PC-2#

PC-2#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/57/64 ms

PC-2#

PC-3#ping 150.150.150.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 150.150.150.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/57/64 ms

PC-3#

PC-3#ping 200.200.200.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 200.200.200.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/59/64 ms

PC-3#

R1#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	100.100.100.3:18	192.168.1.10:18	150.150.150.2:18	150.150.150.2:18
icmp	100.100.100.3:19	192.168.1.10:19	200.200.200.2:19	200.200.200.2:19
icmp	100.100.100.3:6	192.168.1.11:6	150.150.150.2:6	150.150.150.2:6
icmp	100.100.100.3:7	192.168.1.11:7	200.200.200.2:7	200.200.200.2:7
icmp	100.100.100.3:4	192.168.1.12:4	150.150.150.2:4	150.150.150.2:4
icmp	100.100.100.3:5	192.168.1.12:5	200.200.200.2:5	200.200.200.2:5

R1#

R1#sh ip nat statistics

Total active translations: 6 (0 static, 6 dynamic; 6 extended)

Peak translations: 7, occurred 01:35:47 ago

Outside interfaces:

FastEthernet0/1

Inside interfaces:

FastEthernet0/0

Hits: 268 Misses: 0

CEF Translated packets: 268, CEF Punted packets: 0

Expired translations: 19

Dynamic mappings:

-- Inside Source

[Id: 3] access-list 10 pool Public-IP refcount 6

pool Public-IP: netmask 255.255.255.240

start 100.100.100.3 end 100.100.100.3

type generic, total addresses 1, allocated 1 (100%), misses 0

Total doors: 0

Appl doors: 0

Normal doors: 0

Queued Packets: 0

R1#

Google#

*Nov 15 16:34:09.231: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology BASE, dscp 0 topoid 0

*Nov 15 16:34:09.295: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology BASE, dscp 0 topoid 0

*Nov 15 16:34:09.355: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology BASE, dscp 0 topoid 0

*Nov 15 16:34:09.419: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology BASE, dscp 0 topoid 0

*Nov 15 16:34:09.483: ICMP: echo reply sent, src 150.150.150.2, dst 100.100.100.1, topology BASE, dscp 0 topoid 0

Google#

Facebook#

*Nov 15 17:30:33.367: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.3, topology
BASE, dscp 0 topoid 0
 *Nov 15 17:30:33.419: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.3, topology
BASE, dscp 0 topoid 0
 *Nov 15 17:30:33.483: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.3, topology
BASE, dscp 0 topoid 0
 *Nov 15 17:30:33.547: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.3, topology
BASE, dscp 0 topoid 0
 *Nov 15 17:30:33.611: ICMP: echo reply sent, src 200.200.200.2, dst 100.100.100.3, topology
BASE, dscp 0 topoid 0
 Facebook#

R1#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.100.100.1	-	ca07.27a0.0006	ARPA	FastEthernet0/1
Internet	100.100.100.2	36	ca04.2360.0008	ARPA	FastEthernet0/1
Internet	100.100.100.3	-	ca07.27a0.0006	ARPA	FastEthernet0/1
Internet	192.168.1.1	-	ca07.27a0.0008	ARPA	FastEthernet0/0
Internet	192.168.1.10	163	ca01.1f60.0008	ARPA	FastEthernet0/0
Internet	192.168.1.11	131	ca02.1be8.0008	ARPA	FastEthernet0/0
Internet	192.168.1.12	131	ca03.287c.0008	ARPA	FastEthernet0/0

R1#

To Remove NAT overload.

```

R1#config t
R1(config)#no ip nat inside source list 10 pool Public-IP overload
R1(config)#no access-list 10 permit 192.168.1.0 0.0.0.255
R1(config)#no ip nat pool Public-IP 100.100.100.3 100.100.100.3 prefix-length 28
R1(config)#int fa0/0
R1(config-if)#no ip nat inside
R1(config-if)#exit
R1(config)#int fa0/1
R1(config-if)#no ip nat outside
R1(config-if)#exit
R1(config)#exit
R1#
  
```


Important Commands:

```
sh ip nat translations  
sh ip nat statistics  
sh arp
```

Ratan